1 Identifies the three straight edges as $10 + 6 + 10 = 26$ **[M1]**

Calculates the arc length of the semicircle as $\frac{1}{2} \times \pi \times 6$ (or 3π) **[M1]**

$26 + 3\pi$ **[A1]**

[Total Marks: 3]

2 (a) Arc length $BD = \frac{1}{4} \times 2 \times \pi \times 10 = 5\pi$ **[M1]**

Perimeter $= 10 + 10 + 5\pi = 20 + 5\pi$ **[A1]**

(b) Area of quarter circle $= \frac{1}{4} \times \pi \times 10^2 = 25\pi$ **[M1]**

Area of shaded region $= 100 - 25\pi$ **[A1]**

[Total Marks: 4]

3 (a) Curved edge of semi-circle $= \frac{1}{2} \times \pi \times 4 = 2\pi$ **[M1]**

$20 + 2\pi$ **[A1]**

(b) $(8 \times 4) + \left(\frac{1}{2} \times \pi \times 2^2\right)$ **[M1]**

$32 + 2\pi$ **[A1]**

[Total Marks: 4]

4 (a) Perimeter $= 12 + 12 + 8 + \frac{1}{2} \times \pi \times 8$ **[M1]**

$32 + 4\pi$ **[A1]**

(b) Area $= 12 \times 8 + \frac{1}{2} \times \pi \times 4^2$ **[M1]**

$96 + 8\pi$ **[A1]**

[Total Marks: 4]

5 (a) Arc length $= \frac{1}{2} \times \pi \times 8 = 4\pi$ **[M1]**

$32 + 4\pi$ **[A1]**

(b) Area of rectangle $= 12 \times 8 = 96$ **[M1]**

Area of semicircle $= \frac{1}{2} \times \pi \times 4^2 = 8\pi$ **[M1]**

$96 + 8\pi$ **[A1]**

[Total Marks: 5]

6 (a) Area of square $= 4 \times 4 = 16$ **[B1]**

Area of quarter circle $= \frac{1}{4} \times \pi \times 4^2 = 4\pi$ **[M1]**

$16 + 4\pi$ **[A1]**

(b) Arc length $= \frac{1}{4} \times 2 \times \pi \times 4 = 2\pi$ **[M1]**

Perimeter $= 4 + 4 + 4 + 2\pi$ **[M1]**

$12 + 2\pi$ **[A1]**

[Total Marks: 6]

7 (a) Arc length $= \frac{1}{2} \times \pi \times 10$ $(= 5\pi)$ **[M1]**

Perimeter $= 12 + 12 + 10 + 5\pi$ **[M1]**

$34 + 5\pi$ **[A1]**

(b) Area of rectangle $= 12 \times 10$ $(= 120)$ **[M1]**

Area of semicircle $= \frac{1}{2} \times \pi \times 5^2$ $(= 12.5\pi)$ **[M1]**

$120 + 12.5\pi$ **[A1]**

[Total Marks: 6]

8 (a) Calculates the area of the rectangle: $10 \times 6 = 60$ **[M1]**

Calculates the area of the semicircle: $\frac{1}{2} \times \pi \times 3^2$ $(= 4.5\pi)$ **[M1]**

$60 - 4.5\pi$ **[A1]**

(b) Calculates the sum of the three straight edges: $10 + 6 + 10 = 26$ **[M1]**

Calculates the arc length of the semicircle: $\frac{1}{2} \times \pi \times 6$ $(= 3\pi)$ **[M1]**

$26 + 3\pi$ **[A1]**

[Total Marks: 6]

9 (a) Volume of cylinder $= \pi \times 5^2 \times 12 = 300\pi$ **[M1]**

Volume of hemisphere $= \frac{2}{3} \times \pi \times 5^3 = \frac{250\pi}{3}$ **[M1]**

1200 (or 1204 or 1204.2 . . .) **[A1]**

(b) Curved surface of cylinder $= 2 \times \pi \times 5 \times 12 = 120\pi$ **[M1]**

Curved surface of hemisphere + Base of cylinder $= 2 \times \pi \times 5^2 + \pi \times 5^2 = 75\pi$ **[M1]**

613 (or 612.6 . . .) **[A1]**

[Total Marks: 6]

10 (a) Volume of cylinder $= \pi \times 5^2 \times 12$ $(= 300\pi$ or 942.47...) **[M1]**

Volume of cone $= \frac{1}{3} \times \pi \times 5^2 \times 6$ ($= 50\pi$ or 157.07...) **[M1]**

1100 or 1099.5... to 1099.6 **[A1]**

(b) Slant height of cone $= \sqrt{5^2 + 6^2}$ ($= \sqrt{61}$ or 7.81...) **[M1]**

Curved surface area of cone $= \pi \times 5 \times$ their $\sqrt{61}$ (= 122.68...) **[M1]**

Curved surface area of cylinder + base $= (2 \times \pi \times 5 \times 12) + (\pi \times 5^2)$ ($= 145\pi$ or 455.53...) **[M1]**

578 or 578.2... **[A1]**

[Total Marks: 7]

11 (a) Volume of cylinder $= \pi \times 6^2 \times 14$ **[M1]**

Volume of hemisphere $= \frac{2}{3} \times \pi \times 6^3$ **[M1]**

2040 or 2035.7... **[A1]**

(b) Curved surface area of cylinder + base area $= 2 \times \pi \times 6 \times 14 + \pi \times 6^2$ (or $168\pi + 36\pi$) **[M1]**

Curved surface area of hemisphere $= 2 \times \pi \times 6^2$ (or 72π) **[M1]**

Correct addition of all three parts **[M1]**

867 or 867.0... to 867.1 **[A1]**

[Total Marks: 7]

12 (a) Volume of cylinder $= \pi \times 6^2 \times 15 = 540\pi$ **[M1]**

Volume of cone $= \frac{1}{3} \times \pi \times 6^2 \times 8 = 96\pi$ **[M1]**

1998 or 2000 (to 3 sf) **[A1]**

(b) Slant height of cone $l = \sqrt{6^2 + 8^2} = 10$ cm **[M1]**

Curved surface area of cone $= \pi \times 6 \times 10 = 60\pi$ **[M1]**

Curved surface area of cylinder + base area $=$
$2 \times \pi \times 6 \times 15 + \pi \times 6^2 = 180\pi + 36\pi = 216\pi$ **[M1]**

867 (to 3 sf) **[A1]**

[Total Marks: 7]

13 (a) Volume of cylinder $= \pi \times 3^2 \times 10 = 90\pi$ (282.7...) **[M1]**

Volume of hemisphere $= \frac{2}{3} \times \pi \times 3^3 = 18\pi$ (56.5...) **[M1]**

108π or 339 or 339.2 to 339.3 **[A1]**

(b) Area of circular base $= \pi \times 3^2 = 9\pi$ (28.2...) **[B1]**

Curved area of cylinder $= 2 \times \pi \times 3 \times 10 = 60\pi$ (188.4...) **[M1]**

Curved area of hemisphere $= 2 \times \pi \times 3^2 = 18\pi$ (56.5...) **[M1]**

87π or 273 or 273.3 to 273.4 **[A1]**

[Total Marks: 7]

14 **(a)** $\pi \times 4^2 \times 12$ (or 192π) **[M1]**

$\frac{2}{3} \times \pi \times 4^3$ (or $\frac{128\pi}{3}$) **[M1]**

737 or 737.2... or $\frac{704\pi}{3}$ **[A1]**

(b) Calculates curved surface area of cylinder: $2 \times \pi \times 4 \times 12$ (or 96π) **[M1]**

Calculates curved surface area of hemisphere: $2 \times \pi \times 4^2$ (or 32π) **[M1]**

Adds all three external faces including the flat circular base: $96\pi + 32\pi + \pi(4^2)$ **[M1]**

452 or 452.3... to 452.4 or 144π **[A1]**

[Total Marks: 7]

15 **(a)** Volume of large cone $= \frac{1}{3} \times \pi \times 8^2 \times 24$ ($=512\pi$) **[M1]**

Volume of small cone $= \frac{1}{3} \times \pi \times 5^2 \times 15$ ($=125\pi$) **[M1]**

1215.8 (or 1220 or 387π) **[A1]**

(b) Slant height of large cone $= \sqrt{8^2 + 24^2}$ ($=8\sqrt{10}$ or $25.298\ldots$) **[M1]**

Slant height of small cone $= \sqrt{5^2 + 15^2}$ ($=5\sqrt{10}$ or $15.811\ldots$) **[M1]**

Curved SA of frustum $= \pi \times 8 \times 8\sqrt{10} - \pi \times 5 \times 5\sqrt{10}$ ($=387.4\ldots$) **[M1]**

Total SA = Curved SA + $\pi(8^2) + \pi(5^2)$ **[M1]**

667 (or $667.0\ldots$) **[A1]**

[Total Marks: 8]

16 **(a)** Calculates the height of the cylinder: $18 - 5 = 13\,\text{cm}$ **[B1]**

Calculates the volume of the cylinder: $\pi \times 5^2 \times 13$ ($= 325\pi$) **[M1]**

Calculates the volume of the hemisphere: $\frac{2}{3} \times \pi \times 5^3$ ($= \frac{250\pi}{3}$) **[M1]**

1280 (accept 1280 to 1284) **[A1]**

(b) Calculates the curved surface area of the cylinder: $2 \times \pi \times 5 \times 13$ ($= 130\pi$) **[M1]**

Calculates the curved surface area of the hemisphere: $2 \times \pi \times 5^2$ ($= 50\pi$) **[M1]**

Calculates the area of the circular base: $\pi \times 5^2$ ($= 25\pi$) **[M1]**

644 (accept 643 to 645) **[A1]**

[Total Marks: 8]

17 **(a)** Uses similar triangles: $\frac{h}{24} = \frac{4}{10}$ **[M1]**

$h = \dfrac{24 \times 4}{10} = 9.6$ cm **[A1]**

(b) Volume of large cone $= \dfrac{1}{3} \times \pi \times 10^2 \times 24 = 800\pi$ **[M1]**

Volume of small cone $= \dfrac{1}{3} \times \pi \times 4^2 \times 9.6 = 51.2\pi$ **[M1]**

2350 (or 2352.4 . . .) **[A1]**

(c) Sets up equation: $\dfrac{4}{3} \times \pi \times r^3 = 748.8\pi$ (or 2352.4) **[M1]**

$r^3 = 561.6$ **[M1]**

8.25 (or 8.250 . . .) **[A1]**

[Total Marks: 8]

18 (a) Total height of large cone $= 20 + 10 = 30$ cm **[B1]**

Volume of large cone $= \dfrac{1}{3} \times \pi \times 12^2 \times 30$ ($= 1440\pi$ or 4523.89...) **[M1]**

Volume of small cone $= \dfrac{1}{3} \times \pi \times 4^2 \times 10$ ($= 53.33\pi$ or 167.55...) **[M1]**

4360 or 4356... **[A1]**

(b) Slant height of large cone, $L = \sqrt{12^2 + 30^2}$ ($= \sqrt{1044}$ or 32.31...) AND slant height of small cone, $l = \sqrt{4^2 + 10^2}$ ($= \sqrt{116}$ or 10.77...) **[M1]**

Curved surface area of frustum $= \pi \times 12 \times \sqrt{1044} - \pi \times 4 \times \sqrt{116}$ ($= 1082.7...$) **[M1]**

Total surface area $= 1082.7... + \pi \times 12^2 + \pi \times 4^2$ **[M1]**

1585 or 1585.3... to 1585.4 **[A1]**

[Total Marks: 8]

19 (a) Uses similar triangles: $\dfrac{r}{15} = \dfrac{16}{40}$ **[M1]**

$r = 15 \times \dfrac{16}{40} = 6$ leading correctly to given answer **[A1]**

(b) Volume of large cone $= \dfrac{1}{3} \times \pi \times 15^2 \times 40 = 3000\pi$ (9424.7...) **[M1]**

Volume of small cone $= \dfrac{1}{3} \times \pi \times 6^2 \times 16 = 192\pi$ (603.1...) **[M1]**

2808π or 8820 or 8821.5 to 8822 **[A1]**

(c) Equating volumes: $\dfrac{4}{3} \times \pi \times R^3 = 2808\pi$ **[M1]**

$R^3 = 2106$ **[M1]**

$R = \sqrt[3]{2106} = 12.8$ or 12.81 to 12.82 **[A1]**

[Total Marks: 8]

20 (a) Let x be the height of the removed cone. $\dfrac{x}{4} = \dfrac{x + 15}{8}$ **[M1]**

$8x = 4x + 60 \implies 4x = 60 \implies x = 15$. Total height $= 15 + 15 = 30$ cm. **[A1]**

(b) Volume of large cone $= \frac{1}{3} \times \pi \times 8^2 \times 30 = 640\pi$ **[M1]**

Volume of small cone $= \frac{1}{3} \times \pi \times 4^2 \times 15 = 80\pi$ **[M1]**

1760 (to 3 sf) **[A1]**

(c) Slant height of large cone $L = \sqrt{8^2 + 30^2} \approx 31.05$ cm and small cone $l = \sqrt{4^2 + 15^2} \approx 15.52$ cm **[M1]**

Curved surface area of frustum $= \pi \times 8 \times \sqrt{964} - \pi \times 4 \times \sqrt{241} \approx 585.25$ **[M1]**

Area of top and bottom bases $= \pi \times 4^2 + \pi \times 8^2 = 80\pi \approx 251.33$ **[M1]**

837 (to 3 sf) **[A1]**

[Total Marks: 9]

21 (a) Identifies that the centers form an equilateral triangle of side $2r$ **[M1]**

Area of triangle $= \frac{1}{2} \times 2r \times 2r \times \sin(60°) = r^2\sqrt{3}$ **[M1]**

Identifies 3 sectors inside the triangle, each with angle 60°: Total sector area $= 3 \times \frac{60}{360} \times \pi r^2 = \frac{\pi}{2} r^2$ **[M1]**

Area $= r^2\sqrt{3} - \frac{\pi}{2} r^2 = r^2 \left(\sqrt{3} - \frac{\pi}{2}\right)$ with no errors seen **[A1]**

(b) $r^2 \left(\sqrt{3} - \frac{\pi}{2}\right) = 15$ **[M1]**

$r^2 = \frac{15}{\sqrt{3} - \pi/2}$ (≈ 93.02) **[M1]**

9.64 (or 9.644...) **[A1]**

(c) Perimeter consists of 3 arcs, each with angle 60° **[M1]**

Total perimeter $= 3 \times \frac{60}{360} \times 2\pi(9.644\ldots) = \pi(9.644\ldots)$ **[M1]**

30.3 (or 30.29 to 30.30) **[A1]**

[Total Marks: 10]

22 (a) $\frac{r}{10} = \frac{24 - 15}{24}$ or $\frac{r}{10} = \frac{9}{24}$ **[M1]**

3.75 **[A1]**

(b) Volume of large cone $= \frac{1}{3} \times \pi \times 10^2 \times 24$ (or 800π) **[M1]**

Volume of small cone $= \frac{1}{3} \times \pi \times 3.75^2 \times 9$ (or 42.1875π) **[M1]**

2380 or 2381 (accept 2380.7...) **[A1]**

(c) Slant height of large cone $L = \sqrt{10^2 + 24^2} = 26$ **[M1]**

Slant height of small cone $l = \sqrt{3.75^2 + 9^2} = 9.75$ **[M1]**

Curved surface area of frustum $= \pi \times 10 \times 26 - \pi \times 3.75 \times 9.75$ **[M1]**

Adding base and top areas: $+\pi \times 10^2 + \pi \times 3.75^2$ **[M1]**

1060 or 1060.2... to 1060.3 **[A1]**

[Total Marks: 10]

23 **(a)** Uses similar triangles: $\frac{h}{24} = \frac{3}{8}$ **[M1]**

$h = \frac{3 \times 24}{8} = 9$ **[A1]**

(b) Volume of large cone $= \frac{1}{3} \times \pi \times 8^2 \times 24$ (or 512π) **[M1]**

Volume of small cone $= \frac{1}{3} \times \pi \times 3^2 \times 9$ (or 27π) **[M1]**

1520 or 1524 or 485π **[A1]**

(c) Calculates slant height of large cone $L = \sqrt{8^2 + 24^2}$ (or $\sqrt{640}$) and small cone $l = \sqrt{3^2 + 9^2}$ (or $\sqrt{90}$) **[M1]**

Curved surface area of frustum $= \pi(8)(\sqrt{640}) - \pi(3)(\sqrt{90})$ **[M1]**

Calculates area of both flat circular bases: $\pi(8^2) + \pi(3^2)$ (or 73π) **[M1]**

Adds curved surface area and both base areas together **[M1]**

776 (or 775.5... to 775.6) **[A1]**

[Total Marks: 10]

24 **(a)** Sets up an equation using similar triangles (e.g. let x be the height of the smaller cone): $\frac{x}{4} = \frac{x+10}{6}$ (or equivalent) **[M1]**

Solves for the height of the smaller cone: $6x = 4x + 40 \implies 2x = 40 \implies x = 20$ **[M1]**

Calculates the height of the larger cone: $20 + 10 = 30$ **[A1]**

(b) Calculates the volume of the larger cone: $\frac{1}{3} \times \pi \times 6^2 \times 30$ $(= 360\pi)$ **[M1]**

Calculates the volume of the smaller cone: $\frac{1}{3} \times \pi \times 4^2 \times 20$ $(= \frac{320\pi}{3})$ **[M1]**

796 (accept 795 to 797) **[A1]**

(c) Calculates the slant height of the larger cone: $\sqrt{6^2 + 30^2} = \sqrt{936}$ $(= 30.59...)$ **[M1]**

Calculates the slant height of the smaller cone: $\sqrt{4^2 + 20^2} = \sqrt{416}$ $(= 20.39...)$ **[M1]**

Calculates the curved surface area of the frustum: $\pi \times 6 \times \sqrt{936} - \pi \times 4 \times \sqrt{416}$ $(= 320.38...)$ **[M1]**

Calculates the sum of the areas of the circular top and base: $\pi \times 6^2 + \pi \times 4^2$ $(= 52\pi$ or 163.36...) **[M1]**

484 (accept 483.7 to 484.5) **[A1]**

[Total Marks: 11]